

2 SQL Tools and Strategies

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Chapter Overview

- » Setting up our SQL environment
- » The sTunes database
- » DB Browser for SQLite
- » Self-assessments
- » SQL learning strategies

So far, we have talked a bit about what a relational database is, how data is structured inside of a database, and how we can use a relational database management system (RDBMS) to ask questions of the database (by writing queries) and obtain meaningful results. Now that we have gone over the fundamental concepts and terminology, we need to set up a working SQL environment so that we can start practicing queries. This chapter will familiarize you with the resources intended to be used in conjunction with this book, as well as the many exercises and self-assessment questions that will help you test your SQL knowledge.

Introducing the sTunes Database

Imagine you were just hired as an SQL data analyst by an online retail music company, to perform analysis on their music sales data. You are given

access to the company database and told it contains product information (in this case, songs and albums), personal customer information, employee records, and sales data. Management wants to know if the data in the sTunes database contains any useful information about sales, customer demographics, and any ways the company can improve or expand their services. You are given the task of analyzing their database and presenting to

management any insights you discover about the data. You perform all your analysis with SQL. To begin this task, you will need to download a copy of this database onto your computer.

↓ DOWNLOAD

Follow Along and Download the sTunes Company Database

The sample database used throughout the rest of this book is called the sTunes database. Please navigate to www.clydebankmedia.com/sql-assets to access the sTunes database, as well as the free digital assets. After it downloads, do not open it; just save it to a location on your computer and take note of where you save it.

Access the sTunes database and all Digital Asset files for this title at: www.clydebankmedia.com/sql-assets

Introducing DB Browser for SQLite

As we mentioned at the end of chapter 1, we will be working with an RDBMS called SQLite, pronounced "SQL Lite." Although SQLite is the particular implementation or protocol for our sample database, we still need to download an actual application (also known as an SQL browser) that we

can use to "browse" our database just like a web browser is used to interpret pages on the web. SQLite comes with DB Browser, which is a high-quality, visual, open-source tool used to create, design, and edit database files compatible with SQLite. It is for users and developers wanting to create databases, search, and edit data.

Installing DB Browser for SQLite

To install DB Browser, follow the download link included in the digital assets located at www.clydebankmedia.com/sql-assets. Once there, you will find a variety of download links for different operating systems. Be sure to choose the operating system that is relevant to you (32- or 64-bit Windows, Mac OS, Linux, etc.). After downloading the correct file, navigate to where you downloaded it and install the software.

How to Test Your SQL Knowledge

As with any technical skill, practice is the key to mastery. Besides providing examples from a sample database, this book features two types of exercises in the following chapters to assess your newly acquired knowledge

of SQL. Exercises labeled **"ON YOUR OWN"** allow you to quickly practice your skills immediately after a new concept is introduced. As the name implies, these questions do not have written solutions but are not intended to be overly difficult. Another form of self-assessment are the **"Data Analysis Checkpoints."** These exercises are more challenging and may require knowledge of concepts explained in previous chapters. You can find detailed solutions to every Data Analysis Checkpoint in Appendix I.

Strategies for Success

Before we launch our software and open up the sTunes Database, I'd like to offer a few words of advice, to beginner and expert alike. These words come from my experience as a teacher, and following this advice will greatly improve your chances of success:

Type Out Every Code Example by Hand

If you have access to a digital version of this text, do not just copy and paste coding examples into the SQL browser! This advice applies especially to the beginning student who has not experienced SQL before. I am very against the idea that you can learn a programming language by merely copying existing code examples or exercises into the RDBMS environment and observing what they do. Copying and pasting bypasses the syntax, spelling, and muscle memory you activate by rewriting the query yourself. Pasting code from other sources can introduce formatting errors in your code that are difficult to diagnose. For example, pasting code that includes word-processor-formatted quotation marks will often cause a syntax error because DB Browser does not interpret these symbols as single quotation marks. The mistakes we make are just as important as the successes. If the SQL statement you are trying to execute results in a bizarre syntax error, sometimes the best thing to do is delete everything and write the query again from scratch.

Convert SQL Queries to Natural Language Questions

When answering a question or writing a query, try to think of the result as an answer to a real-world question. Instead of thinking "How many entries are in the customers table?" think "How many customers do we have?" This book is intended to help start or enhance your career using

SQL as a tool in your ever-expanding toolbox. The questions you will get from your managers and coworkers will be real-world questions, so it's important to learn the technique of turning a real-world question into an SQL statement and then back into a real-world answer.

Treat This Book Like a Reference Guide

We realize that some of our readers will already have some familiarity with SQL and may want to jump ahead to a particular section. We have designed this book to include the SQL keywords (discussed in each chapter) within the chapter headings so that our readers can easily navigate through the text or quickly go back to review a concept they have already learned. In addition to quick navigation aids, we have included a second appendix with a list of SQL keywords by chapter and some stand-alone examples to get you writing queries as soon as possible. We hope you will keep this book by your desk and refer to it as often as you need to.

CAUTION

Although topic jumping is encouraged, we recommend reading the next chapter (3) first, as it will introduce the software needed to run all of the example queries in the remainder of the book.

Chapter Recap

» This book is designed to introduce the minimum required background information and then jump right in to writing SQL queries.

» To achieve this goal, and to get the most out of this book, it must be used concurrently with database software up and running as well as a sample database.

» As a data analyst, you will often be given an existing database to analyze. The sTunes database file provided here serves as a good example of what an existing database will look like when you first open it.

» DB Browser for SQLite is a free-to-use, public domain database software program that is lightweight and is a good starting point for learning how to examine database files. This program can be run in a Windows or macOS environment.

» Two types of self-assessments are provided in this text to help test your knowledge. The answers to the Data Analysis Checkpoints can be found in Appendix I.

» To make the most of your learning experience, type out the SQL queries in this book by hand (instead of copying them) and consider any SQL question in terms of its real-world

meaning.